24.3.2 Adaptations of Earthworm

It is commonly found in soil, feeding on live and dead organic matter. Earthworm plays a vital role in maintaining soil fertility. It facilitates aeration, water infiltration and producing organic matter to increase crop growth. Some of the adaptations of earthworm are:

Stream-lined body

The earthworm has a cylindrical, elongated and segmented body. This helps them to live in narrow burrows underground and for easy penetration into the soil.

Skin

Mucus covers the skin which does not allow soil particles to stick to it. Moist skin helps in oxygenation of blood.

Burrowing

Its body is flexible having circular and longitudinal muscles which help in movement and subsoil burrowing. Each segment on the lower surface of the body has number of **setae**. They help the earthworm to move through the soil and provide anchor in the burrows.

Aestivation

When the soil becomes too hot or dry, earthworms become inactive and undergo a process called aestivation. Earthworm moves deeper into the soil. It secretes mucus and lowers their metabolic rate in order to reduce water loss. They remain dormant until conditions become favourable. They come out of their burrow during rainy season.

DO YOU KNOW?

Earthworms are referred as '***Farmer's friend***'**.** After digesting organic matter, earthworms excrete a nutrient-rich waste product called castings which is used as Vermicompost.

Nocturnality

Earthworms are sensitive to light. It has no eyes but can sense light through light sensitive cells (**Photoreceptors**) present in their skin. They react negatively to bright light (Photophobic). It remains in its burrow during the day to avoid light.

24.4 Water Conservation

Water conservation is the preservation, control and management of water resources. It also includes activities to protect the hydrosphere and to meet the current and future human demand.

24.4.1 Importance of Water Conservation

- It creates more efficient use of the water resources.
- It ensures that we have enough usable water.
- It helps in decreasing water pollution.
- It helps in increasing energy saving.

24.4.2 Water Conservation Measures

Industrial conservation

Water conservation measures that can be taken by industries are:

- using dry cooling systems.
- if water is used as cooling agent, reusing the water for irrigation or other purposes.

Agricultural conservation

Agricultural water is often lost due to leaks in canals, run off and evaporation. Some of the water conserving methods are:

- using lined or covered canals that reduce loss of water and evaporation.
- using improved techniques such as sprinklers and drip irrigation.
- encouraging the development of crops that require less water and are drought resistant.
- mulching of soil in vegetable cultivation and in horticulture.

DO YOU KNOW?

World Water Day on 22nd March every year, is about focusing attention on the importance of water.

IX_SCI_EM_Unit-24_BIO.indd 296 06-02-2020 13:33:48